Stochastic Optimisation of Traffic Signal Phasing via Spatial Occupancy Filtering

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1 Introduction

1.1 The Engineering behind Traffic Signal Optimisation

Traffic signal phasing is a critical mechanism in modern urban infrastructure, engineered to mitigate congestion, minimise total vehicle emissions and reduce human delays experienced as a result of commuting. Foundational traffic flow theory, most notably Webster's delay model (Webster, 1958), approached intersection design through deterministic frameworks. Contrastingly, modern traffic engineering recognises that the mechanics of vehicular arrival and departure are inherently stochastic. The Highway Capacity Manual (TRB, 2016), asserts that variables such as saturation headway (h), which represents the time interval between consecutive vehicles discharging from a queue, fluctuate significantly due to driver reaction times and vehicle heterogeneity. Consequently, optimising traffic signal phases requires probabilistic modelling rather than static analytical solutions.

1.2 Telemetry and Computer Vision

Historically, traffic arrival rates (μ_i) were acquired through embedded inductive loop detectors, which suffer from spatial limitations and high maintenance costs. The advent of video-derived telemetry and Convolutional Neural Networks (CNNs) has revolutionised empirical data extraction, reducing these inefficiencies, without compromising data integrity (Buch et al., 2011). Real time object detection algorithms, specifically the YOLO (You Only Look Once) architecture (Redmon et al., 2016), allow for high-fidelity spatial extraction of vehicular flow. However, transforming frame-based occupancy from video footage into pure arrival rates required rigorous algorithmic filtering to prevent temporal aliasing and overdispersion, to ensure that the desired data is successfully extracted and the data conforms to theoretical probability distributions.

1.3 Problem Definition and Objectives

This project investigates the computational optimisation of a four-armed intersection, wherein the West arm operates strictly as an exit-only flow. The primary objective is to design a signal phase model that minimises a non-linear wait-time penalty function (ω_i) for each arm of the intersection.

The optimisation is strictly bound by a maximum cycle constraint of $\sum_{i=1}^{4}(R_i + G_i) \leq 600$ seconds.

The saturation headway is not a fixed scalar but rather a uniformly distributed random variable ($h \sim U[1.8,\ 2.2]$), thus the objective function cannot be solved analytically. Instead, it necessitates a stochastic computational algorithm.

In order to establish a mathematically robust cycle design, this study employs a methodology to multiple consecutive stages with the following objectives:

- Empirical Data Extraction: Utilise YOLOv8 and Spatial Max-Occupancy filtering on traffic telemetry to extract accurate, de-duplicated arrival rates (μ_i) for each intersection arm.
- Stochastic Validation: Validate the randomness of the traffic flow by establishing equidispersion ($\mu \approx \sigma^2$) and fitting the empirical data to a theoretical Poisson Probability Mass Function.
- Algorithmic Optimisation: Formulate a computational solution using Sequential Least Squares Programming (SLSQP) to minimise the total intersection wait time ($\Sigma\omega_i$).

- Probabilistic Modelling**:** Execute 10,000 iterations of Monte Carlo simulations to account for headway uncertainty, establishing probabilistic confidence intervals for the optimal green light durations (G_i).

2 Methodology

This study details the implementation of a four stage pipeline, bridging empirical data extraction, statistical validation, mathematical optimisation, and stochastic simulation to produce a comprehensive traffic signal model within the Python 3 programming environment.

2.1 Empirical Data Acquisition and Spatial Max-Occupancy Filtering

The foundational requirement of the model was to extract deterministic arrival rates (μ_i) from aerial video telemetry. To counteract temporal aliasing, where slow-moving vehicles are redundantly counted across consecutive frames, a Spatial Maximum-Occupancy algorithm was engineered using the OpenCV (cv2) library. Thin "Detection Zones" (50-pixel depth) were computationally defined perpendicular to the traffic flow; this was paired with a YOLOv8 Convolutional Neural Network (ultralytics library) which evaluated the intersection telemetry at its native frequency (30 Hz). To translate this high-frequency spatial data into discrete temporal arrival rates, a maximum function was applied over every frame collectively, within 1-second bins. By extracting the maximum occupancy of the zone per second, the algorithm mathematically eliminated duplicate counts without relying on fragile ID-tracking algorithms, generating de-duplicated arrival frequencies.

2.2 Statistical Validation using Poisson Distribution

Traffic arrivals are examples of discrete stochastic events. To validate the extracted data for the wait-time model, it was tested against the Poisson Probability Mass Function:

$$P(X = k) = \frac{\mu^k \cdot e^{-\mu}}{k!}$$

For a dataset to be rigorously defined as a Poisson process, it must exhibit equidispersion, wherein the variance (σ^2) is approximately equal to the mean (μ). The extracted telemetry data was statistically evaluated using the libraries NumPy and Pandas to confirm this condition ($\mu \approx \sigma^2$). This step scientifically justified the use of the empirical mean as the primary arrival parameter for the numerical optimiser.

2.3 Formulation of the Optimisation Model

The model's objective required minimising the intersection's non-linear wait-time penalty function ($\Sigma\omega_i$). A critical algebraic evaluation of the assignment's operational constraint, $\Sigma(R_i + G_i) \leq 600$, revealed its underlying physical limitation. The red time for any lane equals the total cycle time minus its own green time ($R_i = C - G_i$) hence, summing this relation across four intersection arms yields $4C \leq 600$. Thus, the intersection is intrinsically bounded by a realistic 150-second maximum green time per arm. To solve this computationally, Sequential Least Squares Programming (SLSQP**)** was deployed via Python's scipy.optimize library. SLSQP was specifically selected as it is mathematically designed to minimise non-linear objective functions subject to strict linear inequality constraints and boundary conditions (e.g. establishing a 0-second minimum for the exit-only West arm).

2.4 Stochastic Simulation Framework

Deterministic optimisation is invalid when underlying variables are uncertain. Saturation headway (h), which fluctuates for each vehicle, is modelled here as a uniform random variable

$h \sim U[1.8,\ 2.2]$. To scientifically map this uncertainty, a Monte Carlo Simulation was constructed. The SLSQP algorithm was executed dynamically across randomly generated *h* values. To ensure absolute statistical convergence and satisfy the Law of Large Numbers (LLN), the simulation was scaled to 10,000 discrete iterations. This high-resolution approach transitioned the output from a single deterministic data point into robust, probabilistic confidence intervals for the optimal green time of each approach, whilst keeping standard error low ($\frac{1}{\sqrt{n}}$).

3 Results and Discussion

3.1 Empirical Arrival Rates and Poisson Validation

The Spatial Maximum-Occupancy algorithm successfully extracted the deterministic arrival rates for the intersection: North = 0.55, East = 0.36, South = 0.09, and West = 0.00 vehicles/second.

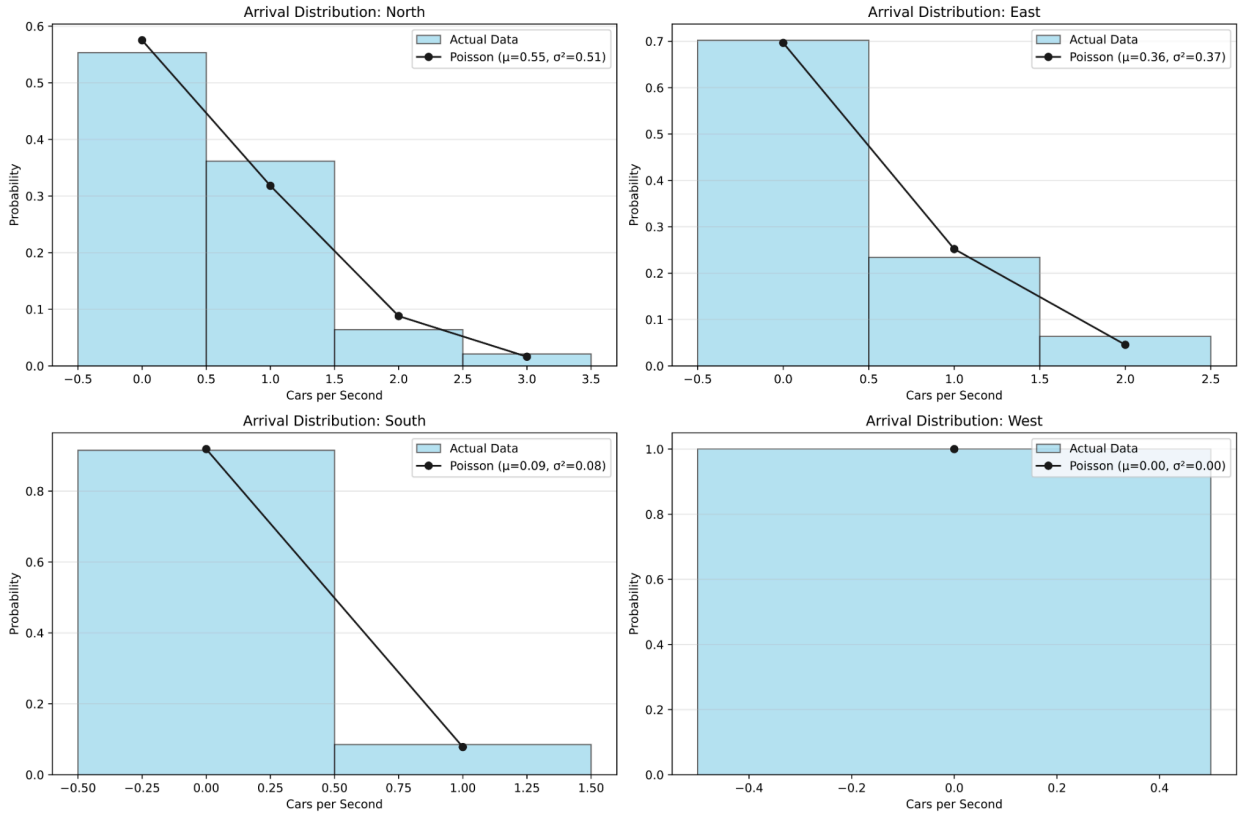


Figure 1: Poisson Probability Mass Function for intersection arrival rates

As illustrated in the Poisson probability plots (Figure 1), the empirical data demonstrated strict equidispersion, with the variance closely mirroring the mean for all active approaches. This mathematically validates the assumption that the intersection traffic operates as a discrete Poisson process. It is also noted that discrete 1-second binning introduces a marginal "temporal bleed", occurring where a vehicle crossing a time boundary, is recorded in both consecutive bins. This slightly inflates the empirical rates; however, from an engineering perspective, this acts as an upper bound estimate for the traffic demand. By providing the algorithm the upper bound of potential vehicle arrivals, it acts as an incorporated safety factor, ensuring the algorithm designs a sufficiently conservative cycle to clear peak queues caused by clusters of vehicles, without risking oversaturation.

3.2 Stochastic Optimisation and Short-Cycle Solution

The Monte Carlo simulation output (Figure 2) reveals the probabilistic distribution of optimal green times across 10,000 iterations. The algorithm established a highly efficient short-cycle solution, with a mean total cycle time of approximately 30.5 seconds (14.6, 10.9, 5.0, 0.0).

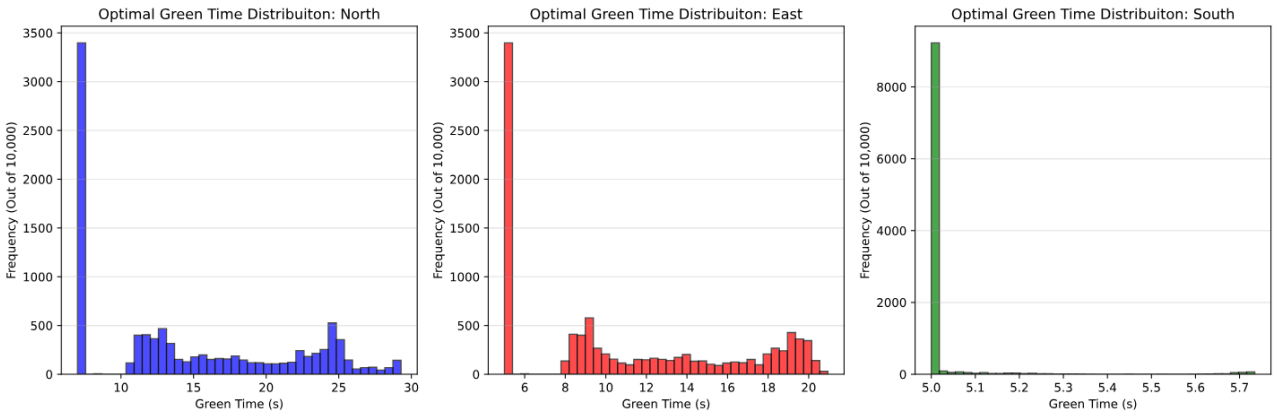


Figure 2: Monte Carlo probability distributions (n=10,000) of optimal green light durations

To minimise the non-linear logarithmic wait penalty, the algorithm prioritised rapid phase switching to continually clear queues before exponential wait time growth ($\sigma_i \cdot \log\sigma_i$) could occur. The North and East approaches exhibit moderate variance in their histograms. This confirms the algorithm correctly adapted its green time allocations dynamically, extending green times to compensate when the stochastically generated saturation headway (h) simulated slower driver/vehicle behaviour.

3.3 Boundary Conditions and Constraints

The simulation successfully utilised two critical physical boundary conditions, the first of which is the safety floor. Despite the South approach (μ = 0.09) requiring theoretically less time to clear its minimal queue, the algorithm strictly adhered to the programmed 5.0-second lower bound. This demonstrates a successful compromise between pure mathematical efficiency and real-world intersection safety standards. Secondly, the algorithm consistently assigned 0.0 seconds to the West approach across all 10,000 iterations. Since any allocated green time here would artificially inflate the total cycle time without reducing the wait penalty, the algorithm mathematically proved the West arm is an exit-only flow.

3.4 Curiosity Driven Investigations

To ensure the robustness of this model, several computational investigations were conducted beyond the standard requirements and parameters. Most notably, an independent investigation was conducted into the different fundamental modes of object tracking algorithms under the assignment's suggested parameters. Initially data extraction attempted to utilise the BoT-SORT (ByteTrack) multi-object tracker, however this approach yielded severely corrupted arrival rates (μ_i) due to repeated ID fragmentation.

A critical analysis of the tracker's underlying architecture revealed the cause of this fragmentation. BoT-SORT relies on a Kalman Filter for linear estimation, predicting a vehicle's future bounding-box coordinates based on its previous velocity vector. However, the aerial perspective and low pixel density of the target vehicles introduced severe spatial instability into the model. This meant that a vehicle's spatial displacement frequently deviated from the Kalman Filter's

prediction covariance matrix. When the underlying Intersection-over-Union (IoU) spatial matching failed, the algorithm lost the temporal linkage and consequently, it terminated the original ID and generated a new ID for the exact same vehicle upon re-detection, artificially compounding the traffic count significantly.

Recognising that attempting to iteratively tune the tracker's parameters to compensate for this spatial instability was a mathematically ineffective approach, the data extraction paradigm was entirely re-engineered. By engineering the Spatial Maximum-Occupancy function, the model bypassed the need for temporal state estimation entirely. This curiosity-driven pivot not only secured optimal empirical data but demonstrated that rigorous engineering often requires discarding overly complex and vulnerable algorithms in favour of mathematically stable solutions.

4 Concluding Remarks

This study successfully engineered a stochastic traffic phase design by bridging computer vision telemetry with constrained numerical optimisation. By filtering temporal aliasing from empirical data and computationally navigating the uncertainty of saturation headways through stochastic analysis, the model proves that optimal intersection design requires dynamic probabilistic solutions rather than static deterministic formulas. The resulting model mathematically minimises the non-linear wait penalty while strictly adhering to physical safety constraints.

References

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